

Information about Individual Contributions

MUST be submitted along with the exam submission

According to Section 15 of the Examination Order, an individual grade or other assessment can only be given in a written group exam if the contribution of each student can be identified (individualisation). If the contribution of each group member is not specified, the exam submission will be rejected.

To assess a group assignment, it is therefore necessary to indicate who is responsible for which sections of your exam submission. This should be specified per section/chapter, not per line. Introduction, partial conclusions, conclusion, and appendices with calculations can be written collaboratively.

Number of characters in each part:

Part 1:	Approx. 12.000
Part 2:	Approx. 12.000
Part 3:	Approx. 11.000

Please indicate the number of students in the group:

3

Please provide information on section responsibility in exam number order:
(If the exam is not anonymous, please provide your name instead of exam number)

Exam number:	Responsible for the sections/chapters:
72	1.1, 1.2.1, 1.2.2, 1.2.4, 1.4.1, 2.3.1, 2.4.2, 5.2.1, 5.2.2, 5.3.3, 5.2.6
282	2.1, 2.5, 1.4.3, 1.4.4, 1.3, 2.3.3, 2.4.3, 5.1.1, 5.2.4
16	1.5, 2.2, 1.2.3, 1.4.2, 2.4.1, 2.3.2, 5.1.2, 5.1.3, 5.1.4, 5.2.5

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☐ **I/we have NOT used generative AI as a tool/aid** (*tick the box*)

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List the generative AI tools used, including a link to the platform (if possible):

ChatGPT <https://chatgpt.com/>

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Briefly describe how generative AI has been used in the assignment (max. 100 words)

GAI has been used to find relevant literature and to correct grammar and spelling errors. ChatGPT has been used as LaTeX-support e.g. to find errors in code or to tell how to insert a footnote in a table.

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Part I: Assignment #2

Relationship Between Inflation and the Interest Rate Spread

1.1 Introduction

This paper investigates the dynamic relationship between the interest rate spread and the inflation rate. The causal contemporaneous relationship of these variables are not theoretically clear. Therefore we utilize a Vector Autoregressive Model framework which does not require us to make assumptions on this causal structure. Using a General To Specific approach and maximum likelihood theory we conduct inference to conduct model selection. This approach along with misspecification tests suggest a reduced VAR(12) model as a reasonable model for the data. This estimated model is stable and well specified. Applying this model, we conduct a joint robust Wald-test we conclude that the data does not indicate Granger causality in either direction.

1.2 Econometric theory

1.2.1 The Vector Auto-Regressive (VAR) Model

We formally define the VAR(k) model by the following stochastic process

$$Z_t = \delta + \Pi_1 Z_{t-1} + \Pi_2 Z_{t-2} + \cdots + \Pi_k Z_{t-k} + \varepsilon_t, \text{ where } \varepsilon_t \sim \text{i.i.d}(0, \Sigma_\varepsilon) \quad (1.1)$$

with $Z_0, Z_{-1}, \dots, Z_{-(k-1)}$ given. Z_t is a $p \times 1$ vector with p variables, Π_i is a $p \times p$ matrix with coefficients on Z_{t-i} (for $i = 1, 2, \dots, k$) and ε_t is the $p \times 1$ vector with stochastic innovations; the error term.

Stability Let $W_t = (Z_t, Z_{t-1}, \dots, Z_{t-k})'$ and the companion matrix C be given by

$$C \equiv \begin{bmatrix} \Pi_1 & \Pi_2 & \cdots & \Pi_{k-1} & \Pi_k \\ I_p & 0 & \cdots & 0 & 0 \\ 0 & I_p & \cdots & 0 & 0 \\ 0 & 0 & \ddots & 0 & 0 \\ 0 & 0 & \cdots & I_p & 0 \end{bmatrix} \quad (1.2)$$

then we can write (1.1) on the *companion form*: $W_t = \tilde{\delta} + CW_{t-1} + \tilde{\varepsilon}_t$ and by the well-known result for the VAR(1) model: The VAR(k) model is stable if the companion matrix C is diagonalizable with all of its eigenvalues inside the complex unit circle.¹

This also implies that the time series processes contained in Z_t are weakly stationary - given that the model correctly specifies the actual data generating process.

¹Theorem 6.2, Nielsen (2025)

1.2.2 (Quasi) Maximum Likelihood Estimation

MLE. This paper estimates the parameters using the Gaussian maximum likelihood estimator defined as

$$\hat{\theta}_{MLE} = \arg \max_{\theta \in \Theta} \log L_T(\theta) \quad (1.3)$$

where Θ is the parameter space and $\log L_T(\theta)$ is the log likelihood function. Formally given by

$$\log L_T(\theta) = \sum_{t=1}^T \ell_t(\theta) = -\frac{Tp}{2} \log(2\pi) - \frac{T}{2} \log(\det(\Sigma)) - \frac{1}{2} \sum_{t=1}^T \epsilon_t(\theta)' \Sigma^{-1} \epsilon_t(\theta) \quad (1.4)$$

, where $\ell_t(\theta)$ is the log likelihood contribution, $\epsilon_t(\theta) = Z_t - \mathbb{E}[Z_t \mid \mathcal{I}_{t-1}]$ and Σ is the variance matrix of $Z_t \in \mathbb{R}^p$. which is also estimated.

It can be shown that if the likelihood function is correctly specified and certain *high level assumptions* hold the MLE is consistent and asymptotically normal distributed such that²

$$\sqrt{T}(\hat{\theta} - \theta_0) \xrightarrow{d} N(0, \Omega_\theta(\theta_0)), \quad T \rightarrow \infty \quad (1.5)$$

Quasi MLE. If the likelihood function is just approximately correctly specified such that the score satisfies $\mathbb{E}[s_t(\theta_0)] = 0$, where $s_t(\theta_0) = \frac{\partial}{\partial \theta} \ell_t(\theta)|_{\theta=\theta_0}$ and if the above mentioned assumptions hold, then the estimator is still asymptotically normal and consistent, but with $\Omega_\theta(\theta_0)$ in (1.5) replaced with the robust variance given by

$$\Omega_\theta^*(\theta_0) = \mathcal{I}(\theta_0)^{-1} \mathcal{J}(\theta_0) \mathcal{I}(\theta_0)^{-1} \quad (1.6^3)$$

In conclusion the QMLE still allows us to conduct inference for hypothesis testing.⁴ Note however, that this estimator is not efficient.

1.2.3 Misspecification tests

To choose the model with the best properties, we test for misspecification using the test listed in table 2. First of all, it is most important that the model has no autocorrelation, as this would imply an inconsistent estimator.

²Nielsen 2025, theorem 3.1

³Where $\mathcal{I}$ is the fisher-information $\mathcal{I}(\theta_0) = -\mathbb{E} \left[\frac{\partial^2 \log \ell_t(\theta_0)}{\partial \theta \partial \theta'} \right]$ and $\mathcal{J}(\theta_0) = \mathbb{E} \left[\frac{\partial \log \ell_t(\theta_0)}{\partial \theta} \frac{\partial \log \ell_t(\theta_0)}{\partial \theta'} \right]$

⁴Nielsen 2025, p. 67

⁵Section 2.3 "Formulation and Misspecification Testing", Nielsen (2025)

Test	Null hypothesis	Asymptotic distribution (under $\mathcal{H}_0$) ^a
Portmanteau test	No autocorrelation	χ^2
No-Heteroskedasticity incl. Cross terms	Unconditional homoscedasticity	χ^2
Normality test	Normally distributed error terms	χ^2
RESET test	Correct functional form	χ^2

^a For more information on degrees of freedom, see Section 2.3 "Formulation and Misspecification Testing," Nielsen (2025).

Table 1: Overview of misspecification tests and their null hypotheses⁵

1.2.4 Granger Causality

Formally, a variable x is said to Granger cause y if:

$$\mathbb{E}[\eta_h^2|\mathcal{I}_t] < \mathbb{E}[\tilde{\eta}_h^2|\mathcal{I}_t] \quad (1.7)$$

holds for some forecast horizon h , where η_h and $\tilde{\eta}_h$ is the forecasting error obtained by forecasting y_{t+h} using the information set $\mathcal{I}_t = \{y_t, x_t, y_{t-1}, x_{t-1}, \dots\}$ and $\tilde{\mathcal{I}}_t = \{y_t, y_{t-1}, \dots\}$ respectively. If this is the case, x is said to *Granger cause* y . This does not refer to causality in a structural or deep naturalistic sense. It limits itself to determining whether x is useful for forecasting y . Note that this implies a temporal condition where the cause comes before the effect.

Test granger causality. Suppose we want to test if $z_i \in Z$ Granger causes $z_j \in Z$ ($i \neq j$), where Z denotes the vector containing all of the endogenous variables. We formalize the null hypothesis by defining the parameter vector $P_{ij} = (\Pi_{ij}^1, \Pi_{ij}^2, \dots, \Pi_{ij}^k)'$ and test

$$\mathcal{H}_0 : P_{ij} = 0 \quad vs. \quad \mathcal{H}_A : P_{ij} \neq 0 \quad (1.8)$$

This jointly tests the statistical significance of lagged variable z_j on predicting z_i . Formally we conduct a Wald test using a robust estimation of the covariance matrix (since all of the models considered indicate heteroscedastic error terms). Under the null, the Wald test is asymptotically $\chi^2(l)$ distributed with l degrees of freedom, where l is the number of restrictions.

1.3 Presentation of the data

In this paper we consider monthly data on the US Consumer Price Index (CPI) and the Interest rate spread. The data is retrieved from FRED and covers a period from 1959(1) to 2025(7).

A visual examination of the two graphs (See Figure 1) reveals that the interest rate spread is cyclical and highly persistent. The characteristics of the CPI is different, with a clear and upwards turning trend. The characteristics of these level series are clearly not in line with weakly stationary processes.

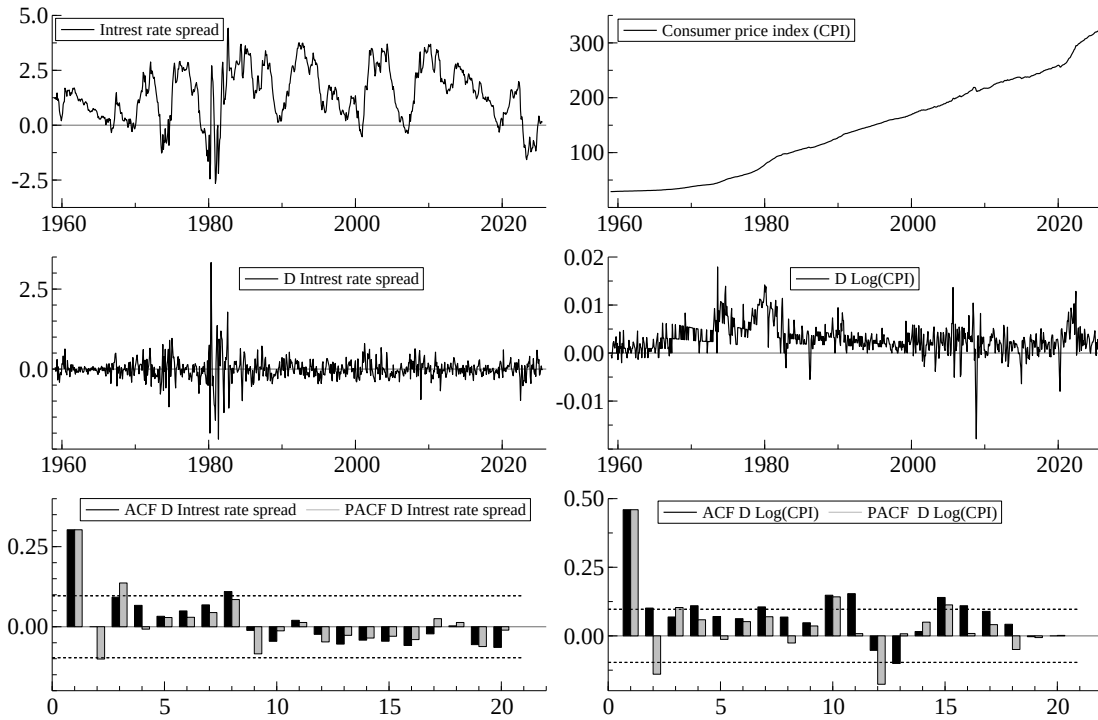


Figure 1: Time series properties

To address the issue of non-stationarity we transform the series to remove trends and unit-root behaviour by taking logs on the CPI and first differences on both series. By graphical inspection of 1 the two series no longer exhibit problematic violations of stationarity. Note, however, that there seems to be some volatility-clustering and outliers. These problems are mostly present in the series before 1990. The rest of the paper will only consider the period from 1990 and onwards. For the above reasons the transformed variables are preferred when modelling the time series.

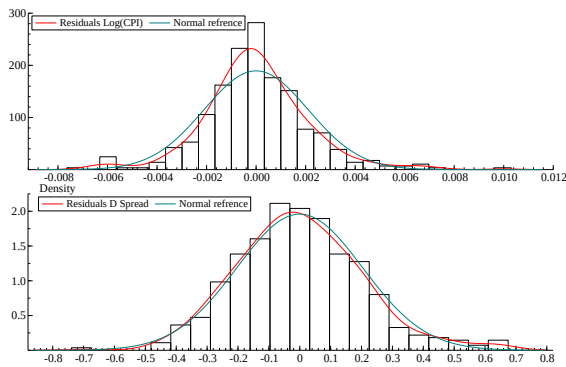
To guide our statistical analysis we compute and plot the (partial) autocorrelation functions (ACF and PACF) from 1990 and onwards (See figure 1). By considering the PACF for the two variables, we find a high level of persistency in the time series. Especially for the inflation rate which show significant partial autocorrelation to about lag 12. However, it is important to note that the PACF is less useful when considering a bivariate VAR model than it is when considering an ARMA model. This is due to the correlations between the two time series. Finally, the empirical tests^{1.4} indicate that the model should allow for around 10-12 lags as a general enough starting point which plausibly nests the true model.

1.4 Empirical results

1.4.1 Model selection

This paper will conduct model selection used the General To Specific (GETS) approach. As mentioned earlier this is motivated by the result that inference is valid if the true DGP is nested as a specific parametrisation of the statistical model, which allows us to test hypothesis on the parameters and reduce the model.

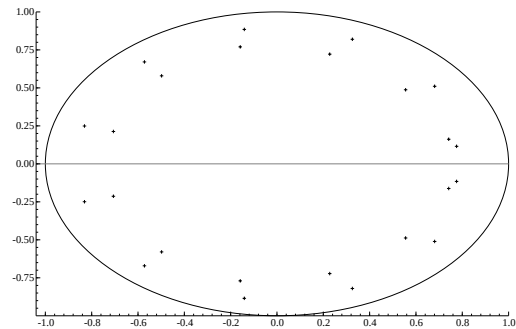
To verify the model specification, we test for autocorrelation, normality, and heteroscedasticity cf. 2.3. Our starting point is the VAR(12) model. This model does not suffer from autocorrelation, but it is strongly rejected that the residuals are normally distributed. Therefore we use QMLE and the robust standard error estimator.



(a) Histogram of residuals

Test	Statistic (p-value)
Portmanteau test	0.98687 [0.4855]
No-Heteroskedasticity	1.8815 [0.0001]
Normality test	72.736 [0.0000]
RESET-test	0.63145 [0.7517]

(b) Misspecification test for the estimated VAR model



(c) Roots of the companion matrix

Figure 2: Misspecification and stability analysis

From here we simplify our model by removing statistically insignificant terms. A joint robust Wald test is conducted with null hypothesis $\mathcal{H}_0 : \Pi_l = 0, l = 4, 5, \dots, 10$. The test results in a p-value of approximately 0.66, which means, we cannot reject the null hypothesis and we omit these lags. We include dummies at critical points, from an economic standpoint, to account for observations that result in large residuals, which we do not believe are generated by a VAR process.

Tabel 2b presents the misspecification tests.⁶ Our estimated model does not indicate autocorrelation or wrong functional form (both the RESET- and Portmantau / Ljung-

⁶Note that the p-values are reliable given that our model is well specified, the series are stationary and certain regularity conditions hold.

Box test have p-values larger than all conventional significant levels). However the test statistics does indicate heteroscedasticity and non-normally distributed errors. However this does not imply that our estimates are inconsistent.

From figure 2c we conclude that the inverse roots of the companion matrix are located within the complex unit circle. This is a sufficient condition for stability and stationarity, cf. subsubsection 1.2.1 given correct model specification.

1.4.2 Estimation results

The estimated parameters of the reduced VAR(12) model for the 427 observations between January 1990 and July 2025 is given in equation (1.9). (Included dummies⁷)

$$\begin{pmatrix} \Delta \text{Spread}_t \\ \Delta \log \text{CPI}_t \end{pmatrix} = \begin{pmatrix} 0.012 (0.016) \\ 0.001300 (0.000211) \end{pmatrix} + \begin{pmatrix} 0.331 (0.050) & 1.873 (4.593) \\ 0.000360 (0.000571) & 0.466 (0.071) \end{pmatrix} \begin{pmatrix} \Delta \text{Spread}_{t-1} \\ \Delta \log \text{CPI}_{t-1} \end{pmatrix} \\ + \begin{pmatrix} -0.120 (0.049) & -5.021 (4.733) \\ -0.000488 (0.000475) & -0.155 (0.066) \end{pmatrix} \begin{pmatrix} \Delta \text{Spread}_{t-2} \\ \Delta \log \text{CPI}_{t-2} \end{pmatrix} + \begin{pmatrix} 0.152 (0.047) & 1.195 (4.558) \\ 0.000432 (0.000486) & 0.097 (0.049) \end{pmatrix} \begin{pmatrix} \Delta \text{Spread}_{t-3} \\ \Delta \log \text{CPI}_{t-3} \end{pmatrix} \\ + \begin{pmatrix} 0.021 (0.052) & -1.945 (4.476) \\ -0.000053 (0.000458) & 0.196 (0.050) \end{pmatrix} \begin{pmatrix} \Delta \text{Spread}_{t-11} \\ \Delta \log \text{CPI}_{t-11} \end{pmatrix} + \begin{pmatrix} -0.036 (0.048) & -0.155 (4.231) \\ 0.0000658 (0.000513) & -0.160 (0.057) \end{pmatrix} \begin{pmatrix} \Delta \text{Spread}_{t-12} \\ \Delta \log \text{CPI}_{t-12} \end{pmatrix} + \begin{pmatrix} u_{1t} \\ u_{2t} \end{pmatrix}. \quad (1.9)$$

The estimated contemporaneous correlation between the two variables are

$$\hat{\rho} = \text{corr}(\Delta \text{Spread}, \Delta \log(\text{CPI})) \approx 0.116 \quad (1.10)$$

We check whether the correlation is significant. Under the null hypothesis, $\hat{\rho}$ is $N(0, T^{-1})$ distributed⁸. The the 5 pct. critical value is $\frac{1.96}{\sqrt{T}} = \frac{1.96}{\sqrt{427}} \approx 0.095$. We reject the null hypothesis, implying that the contemporaneous effect is significantly different from 0 - in line with conventional theory and our expectations regarding the contemporaneous relationship discussed earlier.

The interpretation of (1.10) is that a 1 pct. change in the consumer price index is associated with an increased interest rate spread of 0.116 (corresponding to a relatively higher long rate, or equivalently a relatively lower short rate). This suggests that both series may respond to one another, and may respond contemporaneously to common underlying shocks, such as monetary policy or expectation driven shocks.

⁷The model includes the following dummies: 2008(10), 2008(11), 2008(12), 2022(7). Note that these occur in times of extraordinary insecurity (financial crisis and energy crisis).

⁸(p. 138 Nielsen, 2025)

1.4.3 Granger Causality

We test for Granger Causality between the interest rate spread and the consumer price index using the Wald-test. The null-hypothesis of no-Granger causality is given by

$$\begin{aligned}\mathcal{H}_{\Delta Spread \nrightarrow \Delta \log CPI} : \Pi_{12}^1 &= \Pi_{12}^2 = \Pi_{12}^3 = \Pi_{12}^{11} = \Pi_{12}^{12} = 0 \\ \mathcal{H}_{\Delta \log CPI \nrightarrow \Delta Spread} : \Pi_{21}^1 &= \Pi_{21}^2 = \Pi_{21}^3 = \Pi_{21}^{11} = \Pi_{21}^{12} = 0\end{aligned}\tag{1.11}$$

The corresponding robust Wald-test statistics are presented in (1.12). We cannot reject the null-hypothesis at any conventional level of significance in either case. From this we conclude that neither variable seem to Granger-causes the other at the monthly frequency. In other words, we cannot reject, that past values of the term spread do not help predict inflation, and vice versa.

$$\begin{aligned}W[\Delta Spread \nrightarrow \log(\Delta CPI)] &= 1.7334 \quad (p = 0.8847) \\ W[\log(\Delta CPI) \nrightarrow \Delta Spread] &= 1.7177 \quad (p = 0.8867)\end{aligned}\tag{1.12}$$

1.4.4 Impulse response

To investigate the model's dynamic structure, we calculate the impulse response function (see Figure 3). The impulse response function illustrates how shocks in one variable propagate through its own process over time and affect the other variable in the estimated VAR model.

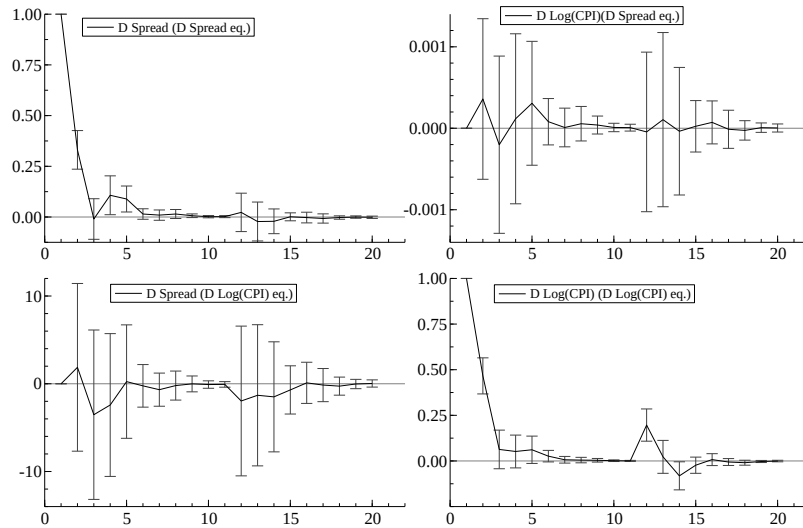


Figure 3: Unit impulse response

From Figure 3 we note that the model suggests, that consumer prices are very weakly affected by shocks to the interest rate spread. There is considerable statistical uncertainty in these estimates; as a result, none of the estimates are significantly different from zero

at the 5% level. Qualitatively the same conclusion holds when observing the effects of shocks to the spread.

However, shocks to each process clearly affect that variable's own dynamic development, especially for the CPI, which strongly depends on its lagged value $\Delta \log(\text{CPIAUCSL}_{t-1})$. Additionally, most of the dynamic effects on the variables themselves dissipate after about three periods (months).

1.5 Conclusion and discussion

In this paper we estimate a reduced VAR(12) model to investigate the dynamic relationship between the interest rate spread and inflation. Our Granger-causality tests provide no evidence that past values of the interest rate spread help predict inflation, nor that past inflation helps predict the spread, at the monthly frequency. However, we cannot reject the presence of contemporaneous dependence between the two series: the estimated contemporaneous correlation is positive (approximately 0.12), indicating that periods with relatively high spreads tend to coincide with periods of relatively high inflation, and vice versa.

In totality, these results suggest that the interest rate spread and inflation do not exert a clear dynamic (lagged) influence on each other in our sample, but they may respond simultaneously to common underlying shocks. An Structural VAR model could be an alternative modelling approach. This however, requires making an assumption on the causal direction between the variables, which we would not advocate in this case. One of the main motivations for using a VAR model is precisely that the contemporaneous causal direction between spread and inflation is unclear.

Overall, we find that the dynamic (lagged) relationship between the variables is weak, even if economic theory might suggest a stronger link. These findings may reflect measurement error or the fact that central banks set short-term rates using information that may not be publicly available at the time of the decision.